

Sukkur Institute of Business Administration University
Department of Computer Science

CS251 – Operating System Lab
BS – V (Software Engineering) – "A"
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Instructor: Sir Abbas Mehdi

Final Project Idea Report

Mini Container Runtime

Submission Profile

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1. Project Name

Mini Container Runtime

2. Group Members (Section A)

1. Muhammad Anas Qadri
2. Masood Ahmed
3. Shaheer Ahmed

3. Brief Project Description

The Mini Container Runtime is a small Linux-based project that demonstrates how operating system features are used to create isolated environments for running programs.

The project will allow users to run a program inside a basic container using a command-line interface. The container will isolate processes from the host system using Linux namespaces and will provide a separate filesystem environment. It will also use Linux cgroups to apply basic resource limits such as memory and the number of processes.

The project will demonstrate important Operating Systems concepts such as process creation, system calls, process isolation, filesystem management, resource management, and process control. The main goal is to understand and demonstrate the basic mechanisms behind container technologies such as Docker.

4. Tools/Technologies to Be Used

- C – Main programming language
- Linux / Ubuntu – Development and execution platform
- GCC – C compiler
- CMake – Build system
- Linux Namespaces – Process, hostname, and mount isolation
- Linux cgroups v2 – Resource management and limits
- Linux system calls – Process and system-level operations
- procfs – Process information inside the container
- Git & GitHub – Version control and team collaboration
- GDB – Debugging
- strace – System call tracing